



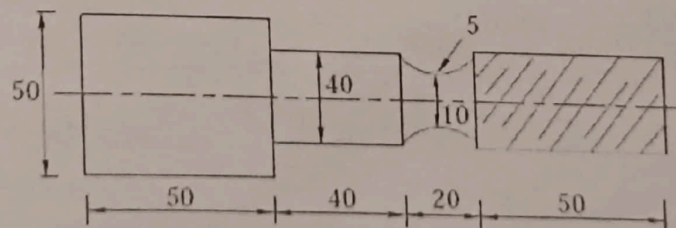
**NATIONAL INSTITUTE OF TECHNOLOGY, TIRUCHIRAPPALLI - 15**  
**DEPARTMENT OF MECHANICAL ENGINEERING**

**End Semester Examination**

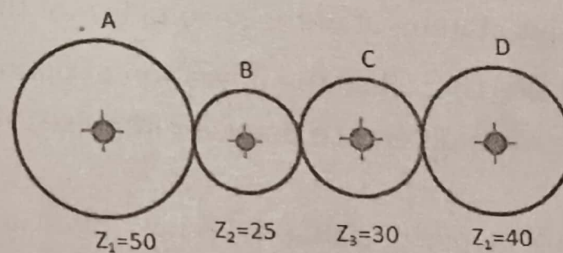
**Programme and specialization** : B.Tech.- Instrumentation and Control Engineering  
**Course code/Title** : MEIR11/Basics of Mechanical Engineering  
**Batch:** July 2025  
**Date:** 30.12.2025  
**Maximum Marks:** 80  
**Time:** 9.30 am to 12.30 pm

Answer all the questions.

1. a) With relevant sketches explain the operation of four stroke Petrol engine. (6) ✓  
b) Write Clausius statement of the second law of thermodynamics. (2)  
c) With a neat sketch explain the operation of open cycle Gas Turbine. (6) ✓  
d) Mention the main difference between Impulse and Reaction turbines. (2)
2. a) Sketch the layout of a thermal power plant and briefly explain the working of its main subsystems. (6) ✓  
b) List out different boiler accessories. (2) ✓  
c) Draw a neat diagram of a storage-type hydroelectric power plant and explain the function of its major components. (6)  
d) Define COP of V.C.Refrigeration system and Ton of refrigeration. (2) ✓
3. a) Describe the casting process, supported by a simple and neat sketch, and explain the function of each stage involved. (10) ✓  
b) List any four major categories of manufacturing processes and give one example for each. (2) ✓  
c) List any two types of milling cutters with simple diagrams. (2) ✓  
d) Write the sequence of machining operations required to manufacture the component shown in the figure from a circular rod of 175 mm length and 60 mm diameter using a lathe machine. (2) ✓



4. a) Classify the metal joining processes and explain any one with neat sketch. (6) ✓  
b) Classify the types of clutches and explain the single-plate clutch with a line diagram. (6) ✓  
c) Write the chemical reactions involved in the generation of nitrogen gas in an automobile airbag system during a collision. (2)  
d) Compute the speed and direction of rotation of the output gear D if the input gear A rotates at 2000 rpm clockwise. The number of teeth on each gear is as shown in the figure. (2)



5. a) Briefly explain any three supporting machine elements with simple sketches. (6)  
b) A solid circular shaft transmits 1.5 MW power at 210 rpm. Determine the diameter of the shaft if the maximum torque exceeds the mean torque by 15%. Take the maximum allowable shear stress as 70 MPa. (6)  
c) A belt drive consists of a 100 mm diameter driver pulley rotating at 1000 rpm, driving a 200 mm diameter driven pulley. Find the speed of the driven pulley. (2)  
d) A gear has a pitch diameter of 100 mm and 50 teeth. Determine the module and circular pitch. (2)